INVESTTECH ENTERPRISE PLATFORM

AI-Powered Investor Intelligence Platform

End-to-End Technical Architecture, Financial KPI Extraction, Real-Time Streaming RAG, and Deployment Specification

Document Version:	v2.4.0 (Production Architecture)
Target Audience:	Executive Stakeholders, Quantitative Analysts, Platform Architects
Underlying LLM Engine:	Azure OpenAI GPT-4o & Local Financial Analyst Reasoning Engine
Vector Search:	Azure AI Search (HNSW Indexing) & Semantic Chunking
Relational Database:	Azure PostgreSQL Flexible Server (Auto SQLite Fallback)
Real-Time Protocol:	Bidirectional WebSockets with Server-Sent Events (SSE) Fallback
Containerization & Cloud:	Docker & Azure Kubernetes Service (AKS)
Ingested Benchmark Filings:	Apple Inc. (AAPL), Microsoft Corp. (MSFT), Tesla Inc. (TSLA) FY2024 Form 10-K

Executive Summary: Modern corporate annual reports (Form 10-K) exceed 100–300 pages of dense disclosures, making rapid cross-company benchmarking and risk identification challenging for investment committees. The InvestTech platform automates the end-to-end extraction lifecycle: turning raw PDF filings into Markdown representations, executing semantic text chunking, generating vector embeddings, extracting audited quantitative KPIs via validated Pydantic schemas, and delivering real-time interactive equity research streaming via WebSockets.

1. System Architecture & Dual-Storage Model

The platform utilizes an asynchronous, decoupled micro-architecture combining a **Dual-Storage Model**. Financial data requires two distinct access patterns: precise, structured numbers for comparative balance sheets and high-dimensional embeddings for qualitative semantic retrieval.

Pillar	Technology	Component Function & Responsibility
Ingestion Engine	PyMuPDF4LLM & LangChain	Converts 10-K PDFs to Markdown, preserving tabular schedules and section hierarchies. Applies semantic chunking based on natural thematic shifts.
Vector Database	Azure AI Search / HNSW	Indexes 1536-dimensional Ada-002 embeddings with metadata filtering (company, year, source_file) for semantic paragraph retrieval.
Relational Store	Azure PostgreSQL / SQLite	Stores structured numerical KPIs in financial_metrics table. Uses SQLAlchemy engine with automatic local SQLite fallback.
RAG & Extraction	Azure OpenAI GPT-4o	Extracts certified KPIs via Pydantic model validation. Powers conversational analyst with retrieved filing context.
Real-Time Streaming	WebSockets & Starlette SSE	Low-latency token streaming, status signaling, cancellation/stop interruption, and ping-pong connection keep-alives.
Web Presentation	FastAPI, Jinja2 & Modern CSS	Executive comparison KPI cards with dynamic bar ratios, qualitative deep-dive inspector, and real-time streaming analyst interface.

2. End-to-End Data Lifecycle & Processing Steps

- **Step 1 — Document Ingestion:** PDF reports uploaded through /api/upload or directory scan are stored in data/raw_pdfs/. Filename metadata is extracted (e.g., 2024_Apple.pdf -> company: Apple, year: 2024).
- **Step 2 — Structural Markdown Parsing:** PDFToMarkdownConverter parses tables into GitHub-flavored Markdown tables and extracts headers, saving the resulting document to data/markdown/.
- **Step 3 — Semantic Chunking:** chunk_markdown() splits markdown documents at natural semantic shift boundaries, preventing cutoffs of financial statement line items.
- **Step 4 — Vector Store Ingestion:** Chunks are converted to embeddings and pushed to Azure AI Search with search attributes (company, year, source_file).
- **Step 5 — Automated KPI Extraction:** kpi_extractor_rag.py executes targeted RAG queries against income statements and balance sheets, validating data into the FinancialMetrics Pydantic schema.
- **Step 6 — Persistent KPI Storage:** Extracted metrics are committed into the financial_metrics relational table via save_metrics().
- **Step 7 — Dashboard & Real-Time Querying:** The executive dashboard reflects new corporate metrics immediately, and the WebSocket AI analyst allows interactive, real-time natural language exploration.

3. Audited FY2024 Corporate Financial Benchmark Data

Below are the verified corporate metrics extracted from the official SEC Form 10-K filings for the three initial platform benchmark enterprises: **Apple Inc.**, **Microsoft Corporation**, and **Tesla, Inc.**

Enterprise	Fiscal Year	Total Revenue	Operating Income	Net Income	Operating Cash Flow	Total Assets	Total Liabilities
Apple Inc. (AAPL)	2024	**$391,035M**	$123,216M	$93,736M	**$118,254M**	$364,980M	$308,030M
Microsoft Corp. (MSFT)	2024	$245,122M	**$109,433M**	**$88,136M**	**$118,548M**	$512,163M	$243,686M
Tesla, Inc. (TSLA)	2024	$97,690M	$7,077M	$7,088M	$13,256M	$114,841M	$46,478M

Calculated Margins & Financial Health Ratios

Company	Operating Margin	Net Profit Margin	Debt-to-Asset Leverage	Cash Conversion (OCF/Net)	Capital Profile
Apple	**31.5%**	24.0%	84.4%	126.2%	High operating leverage, Services expansion, share buybacks
Microsoft	**44.6%** (Leader)	**36.0%** (Leader)	47.6%	134.5%	Industry-leading software margins, Azure hyperscale scaling
Tesla	7.2%	7.3%	**40.5%** (Lowest Debt)	187.0%	Conservative leverage, high capex reinvestment, Energy growth

Executive Equity Research Takeaways:

• **Scale & Top-Line Dominance:** Apple leads the cohort with $391,035M in revenue, backed by its active base of 2.2B+ devices and $96B+ in high-margin Services revenue.

• **Operating Leverage & Pricing Power:** Microsoft generates the highest profitability with a 44.6% operating margin and 36.0% net margin, driven by commercial cloud and Microsoft 365 Copilot monetization.

• **Cash Generation:** Apple and Microsoft each generated over $118 Billion in operating cash flows in 2024. Tesla generated $13,256M in cash flow while funding intensive manufacturing expansion and Dojo AI compute clusters.

4. Qualitative Intelligence: Growth Drivers vs. 10-K Risks

Beyond raw balance sheets, Form 10-K Item 1A (Risk Factors) and Item 7 (MD&A;) disclose forward-looking catalysts and existential business threats. The platform automatically extracts and presents these qualitative factors:

Apple Inc. (AAPL) — Strategic Catalysts & Principal Headwinds

Top Growth Drivers	Principal 10-K Risk Disclosures
• Record expansion in Services revenue fueled by iCloud, App Store, Apple Pay, and subscription platforms. • Global demand for iPhone 16 series powered by Apple Intelligence on-device neural architecture. • Mac and iPad ecosystem momentum driven by next-generation M-series Apple Silicon efficiency. • Active installed device base surpassing 2.2 billion active devices worldwide. • Disciplined capital allocation with continuous share repurchase programs ($95B+ returned).	• Macroeconomic conditions including inflation, interest rates, and foreign exchange fluctuations impacting consumer electronics demand. • Intense global smartphone competition with aggressive pricing and compressed product refresh cycles. • Dependence on single- or limited-source manufacturing partners (e.g., TSMC, Foxconn), creating supply vulnerability. • Regulatory, antitrust, and tax investigations worldwide, including European Union Digital Markets Act (DMA) compliance. • Geopolitical exposure in Greater China representing a critical manufacturing base and major sales market.

Microsoft Corporation (MSFT) — Strategic Catalysts & Principal Headwinds

Top Growth Drivers	Principal 10-K Risk Disclosures
• Exceptional growth across Microsoft Cloud and Azure OpenAI enterprise infrastructure adoption. • Rapid enterprise monetization of Microsoft 365 Copilot, Security Copilot, and GitHub Copilot. • Hybrid cloud expansion across Server products and multi-cloud enterprise deployments. • Integration of Activision Blizzard significantly expanding gaming and Xbox Game Pass content monetization. • Commercial cloud gross margin resilience supported by hyperscale datacenter operational efficiencies.	• Substantial capital expenditures required to expand global datacenter and specialized AI computing clusters ($50B+ Capex). • Evolving global regulations and legal scrutiny surrounding artificial intelligence and algorithmic accountability. • Sophisticated cybersecurity threats and nation-state attacks targeting cloud software services. • Intense cloud competition from Amazon AWS and Google Cloud Platform in AI infrastructure. • Geopolitical restrictions on international technology transfers and advanced semiconductor availability.

Tesla, Inc. (TSLA) — Strategic Catalysts & Principal Headwinds

Top Growth Drivers	Principal 10-K Risk Disclosures
• Record expansion in Tesla Energy segment with record Megapack and Powerwall deployment volume (125%+ YoY). • Continued vehicle delivery leadership of Model Y as the world's best-selling passenger vehicle. • Rapid technological advancements in Full Self-Driving (Supervised) end-to-end neural network models. • Ongoing development of next-generation affordable vehicle platforms and Cybercab autonomous robotaxi fleet. • Global adoption of the North American Charging Standard (NACS) and Supercharger licensing agreements.	• Intensifying pricing pressure and aggressive competition across global electric vehicle markets, notably from Chinese OEMs. • Technological and regulatory uncertainty regarding autonomous vehicle validation and commercial robotaxi operation. • Volatility in critical battery raw material costs including lithium, nickel, and refined graphite. • Production scaling hurdles for new vehicle architectures including Cybertruck and 4680 dry-electrode cells. • Key-person dependency on executive leadership and elevated brand sensitivity.

5. Real-Time Streaming Chatbot Architecture

A standout innovation of the platform is the **real-time streaming conversational agent**. Unlike standard chatbots that freeze for several seconds while awaiting a complete payload, this engine streams response tokens and progress status events dynamically.

Direction	Event Type	Payload Example	System Behavior
Client -> Server	chat	`{'question': '...', 'company': 'Apple', 'year': 2024}`	Initiates retrieval and starts asynchronous response token streaming.
Server -> Client	status	`{'message': 'Analyzing 10-K filings...'}`	Updates the client status indicator with live retrieval feedback.
Server -> Client	chunk	`{'text': '### Revenue Analysis '}`	Appends token chunk into active message bubble with blinking cursor.
Client -> Server	stop	`{'type': 'stop'}`	Cancels the active streaming asyncio Task cleanly on demand.
Server -> Client	done	`{'full_text': '...'}`	Removes streaming cursor, binds Copy and Text-to-Speech action tools.
Bidirectional	ping/pong	`{'type': 'ping'} -> {'type': 'pong'}`	Heartbeat every 25 seconds keeps connection active through firewalls.

6. Platform API Reference & Endpoint Specifications

FastAPI delivers automated OpenAPI documentation (Swagger UI accessible at http://localhost:8000/docs). The primary endpoints are specified below:

Method	Endpoint Path	Parameters / Payload	Response Output & Description
GET	`/`	None	Renders dashboard UI with company KPI cards, comparison bars, and chat panel.
GET	`/health`	None	Returns {'status': 'healthy'} for Kubernetes liveness/readiness probes.
GET	`/api/metrics`	None	Returns raw JSON array of all stored audited company financial metrics and ratios.
POST	`/api/upload`	multipart/form-data (file: PDF)	Uploads PDF report, runs Markdown conversion, chunking, and KPI extraction.
WS	`/ws/chat`	WebSocket JSON messages	Bidirectional real-time stream for interactive financial analyst conversation.
POST	`/api/chat/stream`	`{'question': str, 'company': str}`	Server-Sent Events (SSE) streaming fallback endpoint.
POST	`/api/chat`	`{'question': str, 'company': str}`	Synchronous non-streaming JSON response endpoint for legacy integrations.

7. Cloud Deployment & Container Orchestration

The platform is built as a cloud-native microservice ready for Azure Kubernetes Service (AKS) deployment:

- **Containerization (Docker):** Multi-stage Python 3.12-slim base image using Astral UV for fast dependency installation. Exposes port 8000 and runs Uvicorn ASGI server.
- **Container Registry (ACR):** Docker images are tagged and pushed to Azure Container Registry (myacr.azurecr.io/investor-intelligence:latest).
- **Kubernetes Manifests (k8s/):** deployment.yaml provisions pod replicas with CPU/memory limits, liveness/readiness probes to /health, and mounts Azure Key Vault secrets for database credentials.
- **Automated Provisioning (setup_azure.ps1):** PowerShell script provisions resource groups, Azure PostgreSQL Flexible Server, Azure AI Search instance, Azure OpenAI deployments, and AKS clusters in a single command.
- **Resilient Local Fallback:** Zero external lock-in. If Azure credentials are not provided, the platform seamlessly switches to local SQLite storage and internal 10-K filing search, guaranteeing uninterrupted operation.

Document Status: Verified & Finalized | InvestTech Platform Architecture